

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

Homework08 - MethodDefinitionOnChickenFarmApp

LASCANO, JORGE EDISON

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ESPE 2609 – OBJECT-ORIENTED PROGRAMMING SW 36863

Student: Dylan Cumbajin
Course: Object-Oriented Programming
Homework: 08
Partial: 1

Topic: Implementation of ChickenCoop Methods

Function 1: add()

```java
public void add(Chicken chicken){
    // algorithn
    // If have less than 10 chickens, we can add one more
    //else we can not add any more
    if (chicken == null) {
        return;
    }

    for (int i = 0; i < chickens.length; i++) {
        if (chickens[i] == null) {
            chickens[i] = chicken;
            System.out.println("Chicken added --> " + chicken.getName());
            return;
        }
    }
    System.out.println("The chicken coop is full");
}
```

Function 2: remove()

```java
public void remove(int chickenId){
    for (int i = 0; i < chickens.length; i++) {
        if (chickens[i] != null && chickens[i].getId() == chickenId) {

            for (int j = i; j < chickens.length - 1; j++) {
                chickens[j] = chickens[j + 1];
            }

            chickens[chickens.length - 1] = null;

            if (currentIndex > i) {
                currentIndex--;
            }

            System.out.println("Chicken removed --> " + chickenId);
            return;
        }
    }
    System.out.println("Chicken not found");
}
```

Function 3: next()

```java
public Chicken next(){
    while (currentIndex < chickens.length) {
        Chicken chicken = chickens[currentIndex];
        currentIndex++;

        if (chicken != null) {
        return chicken;
        }
    }
    return null;
}
```

Execution Results

```
Output - ChickenFarmApp (run)
chickens --> Chicken{id=8, name=chicken 9, color=white, bornOnDate=Tue Apr 03 00:00:00 GMT-05:00 2018, ageInYears
chickens --> Chicken{id=9, name=chicken 10, color=white, bornOnDate=Wed Apr 03 00:00:00 GMT-05:00 2019, ageInYear

--- ADD CHICKENS ---
Chicken added --> Maruja
Chicken added --> Lucy

--- LIST CHICKENS ---
Chicken --> Maruja, ID --> 2
Chicken --> Lucy, ID --> 1

--- REMOVE CHICKEN ---
Chicken removed --> 2

--- REMAINING CHICKENS ---
Chicken --> Lucy, ID --> 1
BUILD SUCCESSFUL (total time: 1 second)
```